Lec 06 Stochastic Approximation and Stochastic Gradient Descent

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Motivating example: mean estimation

🏷️ **Revisit the mean estimation problem:**

- Consider a random variable X.
- Our aim is to estimate $\mathbb{E}[X]$.
- Suppose that we collected a sequence of **iid** samples $\{x_i\}_{i=1}^N$.
- Then the expectation of X can be approximated by

$$\mathbb{E}[X] \approx \bar{x} := \frac{1}{N}\sum_{i=1}^{N} x_i.$$

Incremental mean estimation

In particular, suppose

$$w_{k+1} = \frac{1}{k}\sum_{i=1}^{k} x_i, \quad k = 1, 2, \ldots$$

and hence

$$w_k = \frac{1}{k-1}\sum_{i=1}^{k-1} x_i, \quad k = 2, 3, \ldots$$

Then, w_{k+1} can be expressed in terms of w_k as

$$\begin{aligned} w_{k+1} = \frac{1}{k}\sum_{i=1}^{k} x_i &= \frac{1}{k}(\sum_{i=1}^{k-1} x_i + x_k) \\ &= \frac{1}{k}((k-1)w_k + x_k) = w_k - \frac{1}{k}(w_k - x_k). \end{aligned}$$

Therefore, we obtain the following iterative algorithm:

$$w_{k+1} = w_k - \frac{1}{k}(w_k - x_k).$$

Furthermore, consider an algorithm with a more general expression:

$$w_{k+1} = w_k - \alpha_k(w_k - x_k),$$

where $1/k$ is replaced by $\alpha_k > 0$.

- It can be proved that the algorithm still converge to the mean $\mathbb{E}[X]$ if $\{\alpha_k\}$ satisfy some mild conditions.
- This algorithm is a special **SA algorithm** and also a special **stochastic gradient descent algorithm**.

RM algorithm

Problem statement

Suppose we would like to find the root of the equation

$$g(w) = 0,$$

where $w \in \mathbb{R}$ is the variable to be solved and $g : \mathbb{R} \to \mathbb{R}$ is a function.

The algorithm

The **Robbins-Monro (RM) algorithm** that can solve this problem is as follows:

$$w_{k+1} = w_k - a_k \tilde{g}(w_k, \eta_k), \quad k = 1, 2, 3, \ldots$$

where

- w_k is the kth estimate of the root.
- $\tilde{g}(w_k, \eta_k) = g(w_k) + \eta_k$ is the k th noisy observation.
- a_k is a positive coefficient.

The function $g(w)$ is viewed as a black box, for which only the input sequence $\{w_k\}$ and output sequence (noisy) $\{\tilde{g}(w_k, \eta_k)\}$ are available. So this algorithm relies on data instead of model.

Convergence Analysis

Theorem:

Theorem (Robbins-Monro Theorem)

In the Robbins-Monro algorithm, if

1) $0 < c_1 \leq \nabla_w g(w) \leq c_2$ *for all* w;
2) $\sum_{k=1}^{\infty} a_k = \infty$ *and* $\sum_{k=1}^{\infty} a_k^2 < \infty$;
3) $\mathbb{E}[\eta_k | \mathcal{H}_k] = 0$ *and* $\mathbb{E}[\eta_k^2 | \mathcal{H}_k] < \infty$;

where $\mathcal{H}_k = \{w_k, w_{k-1}, \ldots\}$, *then* w_k *converges with probability 1 (w.p.1) to the root* w^* *satisfying* $g(w^*) = 0$.

Explanations of the three conditions:

- **Condition 1:**

- g should be monotonically increasing, which ensures that the root of $g(w) = 0$ exists and is unique.

- **Condition 2:**
  - $\sum_{k=1}^{\infty} a_k^2 < \infty$ ensures that a_k converges to zero as $k \to \infty$,so w_k converges to w^* as well.
  - $\sum_{k=1}^{\infty} a_k = \infty$ ensures that a_k do not converge to zero too fast. Otherwise a_k might converge to zero too early when there is still quite a distance between w_k and w^*, and in this case w_k is not able to get closer to w^*.

- **Condition 3:**
  - special yet common case is that $\{\eta_k\}$ is an iid stochastic sequence satisfying $\mathbb{E}[\eta_k] = 0$ and $\mathbb{E}[\eta_k^2] < \infty$. The observation error η_k is not required to be Gaussian.

Examine the second condition more closely:

- $\sum_{k=1}^{\infty} a_k^2 < \infty$ indicates that a_k converges to zero as $k \to \infty$.
  Since

$$w_{k+1} - w_k = -a_k \tilde{g}(w_k, \eta_k),$$

  - If $a_k \to 0$, then $a_k \tilde{g}(w_k, \eta_k) \to 0$ and hence $w_{k+1} - w_k \to 0$.
  - We need the fact that $w_{k+1} - w_k \to 0$ if w_k converges eventually.
  - If $w_k \to w^*$,$g(w_k) \to 0$ and $\tilde{g}(w_k, \eta_k)$ is dominant by η_k.

- $\sum_{k=1}^{\infty} a_k = \infty$ indicates that a_k do not converge to zero too fast.
  Summarizing $w_2 = w_1 - a_1 \tilde{g}(w_1, \eta_1)$,$w_3 = w_2 - a_2 \tilde{g}(w_2, \eta_2)$, $\ldots, w_{k+1} = w_k - a_k \tilde{g}(w_k, \eta_k)$leads to

$$w_1 - w_\infty = \sum_{k=1}^{\infty} a_k \tilde{g}(w_k, \eta_k).$$

  Suppose $w_\infty = w^*$. If $\sum_{k=1}^{\infty} a_k < \infty$, then$\sum_{k=1}^{\infty} a_k \tilde{g}(w_k, \eta_k)$ may be bounded. Then, if the initial guess $w_1$is chosen arbitrarily far away from w^*, then the above equality would be invalid.

Apply to mean estimation

Proof: mean estimation is a special case of RM algorithm

1) Consider a function:

$$g(w) \doteq w - \mathbb{E}[X].$$

Our aim is to solve $g(w) = 0$. If we can do that, then we can obtain $\mathbb{E}[X]$.

2) The observation we can get is

$$\tilde{g}(w, x) \doteq w - x,$$

because we can only obtain samples of X. Note that

$$\begin{aligned}\tilde{g}(w, \eta) = w - x &= w - x + \mathbb{E}[X] - \mathbb{E}[X] \\ &= (w - \mathbb{E}[X]) + (\mathbb{E}[X] - x) \doteq g(w) + \eta,\end{aligned}$$

3) The RM algorithm for solving $g(x) = 0$ is

$$w_{k+1} = w_k - \alpha_k \tilde{g}(w_k, \eta_k) = w_k - \alpha_k (w_k - x_k),$$

which is exactly the mean estimation algorithm.

The convergence naturally follows.

SGD

Introduction

Suppose we aim to solve the following optimization problem:

$$\min_w \quad J(w) = \mathbb{E}[f(w, X)]$$

- w is the parameter to be optimized.
- X is a random variable. The expectation is with respect to X.
- w and X can be either scalars or vectors. The function $f(\cdot)$ is a scalar.

Method 1: gradient descent (GD)

$$w_{k+1} = w_k - \alpha_k \nabla_w \mathbb{E}[f(w_k, X)] = w_k - \alpha_k \mathbb{E}[\nabla_w f(w_k, X)]$$

Drawback: the expected value is difficult to obtain.

Method 2: batch gradient descent (BGD)

$$\mathbb{E}[\nabla_w f(w_k, X)] \approx \frac{1}{n}\sum_{i=1}^{n} \nabla_w f(w_k, x_i).$$

$$w_{k+1} = w_k - \alpha_k \frac{1}{n}\sum_{i=1}^{n} \nabla_w f(w_k, x_i).$$

Drawback: it requires many samples in each iteration for each w_k.

Method 3: stochastic gradient descent (SGD)

$$w_{k+1} = w_k - \alpha_k \nabla_w f(w_k, x_k),$$

- Compared to the gradient descent method: Replace the **true gradient** $\mathbb{E}[\nabla_w f(w_k, X)]$ by the **stochastic gradient** $\nabla_w f(w_k, x_k)$.
- Compared to the batch gradient descent method: let $n = 1$.

Convergence

From GD to SGD:

$$\begin{gathered} w_{k+1} = w_k - \alpha_k \mathbb{E}[\nabla_w f(w_k, X)] \\ \Downarrow \\ w_{k+1} = w_k - \alpha_k \nabla_w f(w_k, x_k) \end{gathered}$$

$\nabla_w f(w_k, x_k)$ can be viewed as a noisy measurement of $\mathbb{E}[\nabla_w f(w, X)]$:

$$\nabla_w f(w_k, x_k) = \mathbb{E}[\nabla_w f(w, X)] + \underbrace{\nabla_w f(w_k, x_k) - \mathbb{E}[\nabla_w f(w, X)]}_{\eta}.$$

Since

$$\nabla_w f(w_k, x_k) \neq \mathbb{E}[\nabla_w f(w, X)]$$

Now we show that **SGD is a special RM algorithm**. Then, the convergence naturally follows.

The aim of SGD is to minimize

$$J(w) = \mathbb{E}[f(w, X)]$$

This problem can be converted to a root-finding problem:

$$\nabla_w J(w) = \mathbb{E}[\nabla_w f(w, X)] = 0$$

Let

$$g(w) = \nabla_w J(w) = \mathbb{E}[\nabla_w f(w, X)].$$

Then, the aim of SGD is to find the root of $g(w) = 0$.

What we can measure is

$$\begin{aligned}\tilde{g}(w, \eta) &= \nabla_w f(w, x) \\ &= \underbrace{\mathbb{E}[\nabla_w f(w, X)]}g(w) + \underbrace{\nabla_w f(w, x) - \mathbb{E}[\nabla_w f(w, X)]}\eta.\end{aligned}$$

Then, the RM algorithm for solving $g(w) = 0$ is

$$w_{k+1} = w_k - a_k \tilde{g}(w_k, \eta_k) = w_k - a_k \nabla_w f(w_k, x_k).$$

- It is exactly the SGD algorithm.
- Therefore, SGD is a special RM algorithm.

🏷️ **Theorem:**

Theorem (Convergence of SGD)

In the SGD algorithm, if

1) $0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$;
2) $\sum_{k=1}^{\infty} a_k = \infty$ *and* $\sum_{k=1}^{\infty} a_k^2 < \infty$;
3) $\{x_k\}_{k=1}^{\infty}$ *is iid;*

then w_k *converges to the root of* $\nabla_w \mathbb{E}[f(w, X)] = 0$ *with probability 1.*

BGD, MBGD, and SGD

Suppose we would like to minimize $J(w) = \mathbb{E}[f(w, X)]$ given a set of random samples $\{x_i\}_{i=1}^n$ of X. The BGD, SGD, MBGD algorithms solving this problem are, respectively,

$$w_{k+1} = w_k - \alpha_k \frac{1}{n} \sum_{i=1}^{n} \nabla_w f(w_k, x_i), \qquad \text{(BGD)}$$

$$w_{k+1} = w_k - \alpha_k \frac{1}{m} \sum_{j \in \mathcal{I}_k} \nabla_w f(w_k, x_j), \qquad \text{(MBGD)}$$

$$w_{k+1} = w_k - \alpha_k \nabla_w f(w_k, x_k). \qquad \text{(SGD)}$$

- **In the BGD algorithm**, all the samples are used in every iteration. When n is large, $(1/n)\sum_{i=1}^n \nabla_w f(w_k, x_i)$ is close to the true gradient $\mathbb{E}[\nabla_w f(w_k, X)]$.
- **In the MBGD algorithm**, $\mathcal{I}_k$ is a subset of $\{1, \ldots, n\}$ with the size as $|\mathcal{I}_k| = m$. The set $\mathcal{I}_k$ is obtained by m times idd samplings.
- **In the SGD algorithm**, x_k is randomly sampled from $\{x_i\}_{i=1}^n$ at time k.

Compare MBGD with BGD and SGD:

- Compared to SGD, MBGD has less randomness because it uses more samples instead of just one as in SGD.
- Compared to BGD, MBGD does not require to use all the samples in every iteration, making it more flexible and efficient.
- If $m = 1$, MBGD becomes SGD.
- If $m = n$, MBGD does NOT become BGD strictly speaking because MBGD uses randomly fetched n samples whereas BGD uses all n numbers. In particular, MBGD may use a value in $\{x_i\}_{i=1}^n$ multiple times whereas BGD uses each number once.